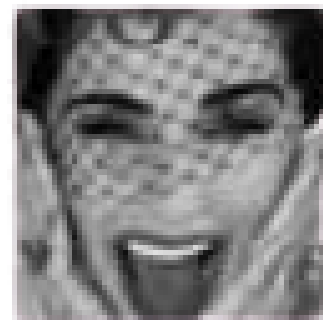
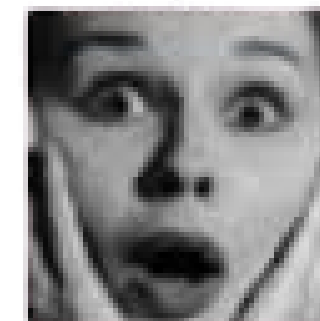




EE4016 Group Project

Group 10

MoodMirror



KAPYA Zachariah Muya (Team Leader)

WANG Xinan

Cheng Wing Yan

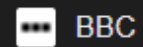
YIP Chung Hon

Sit Yan Tung

Communication is more than words



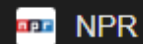
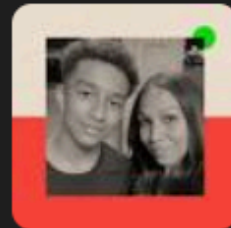
Applications



'A predator in your home': Mothers say chatbots encouraged their sons to kill themselves

In her first UK interview Megan Garcia speaks to Laura Kuenssberg about the death of her teenage son.

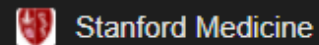
8 Nov 2025



Their teenage sons died by suicide. Now, they are sounding an alarm about AI chatbots

Grieving parents and online safety advocates at a congressional hearing called for new laws to regulate AI companion apps to protect the...

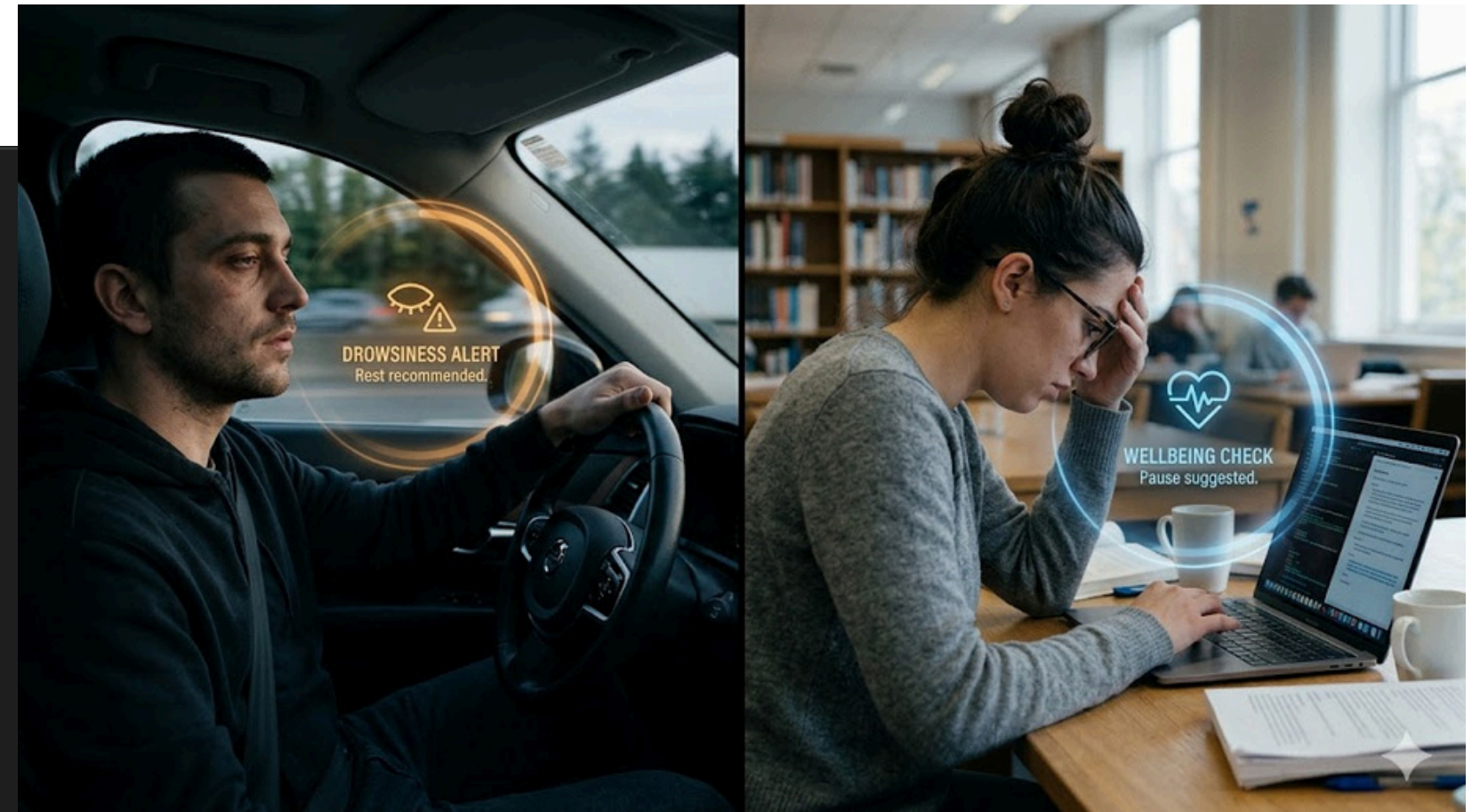
19 Sept 2025



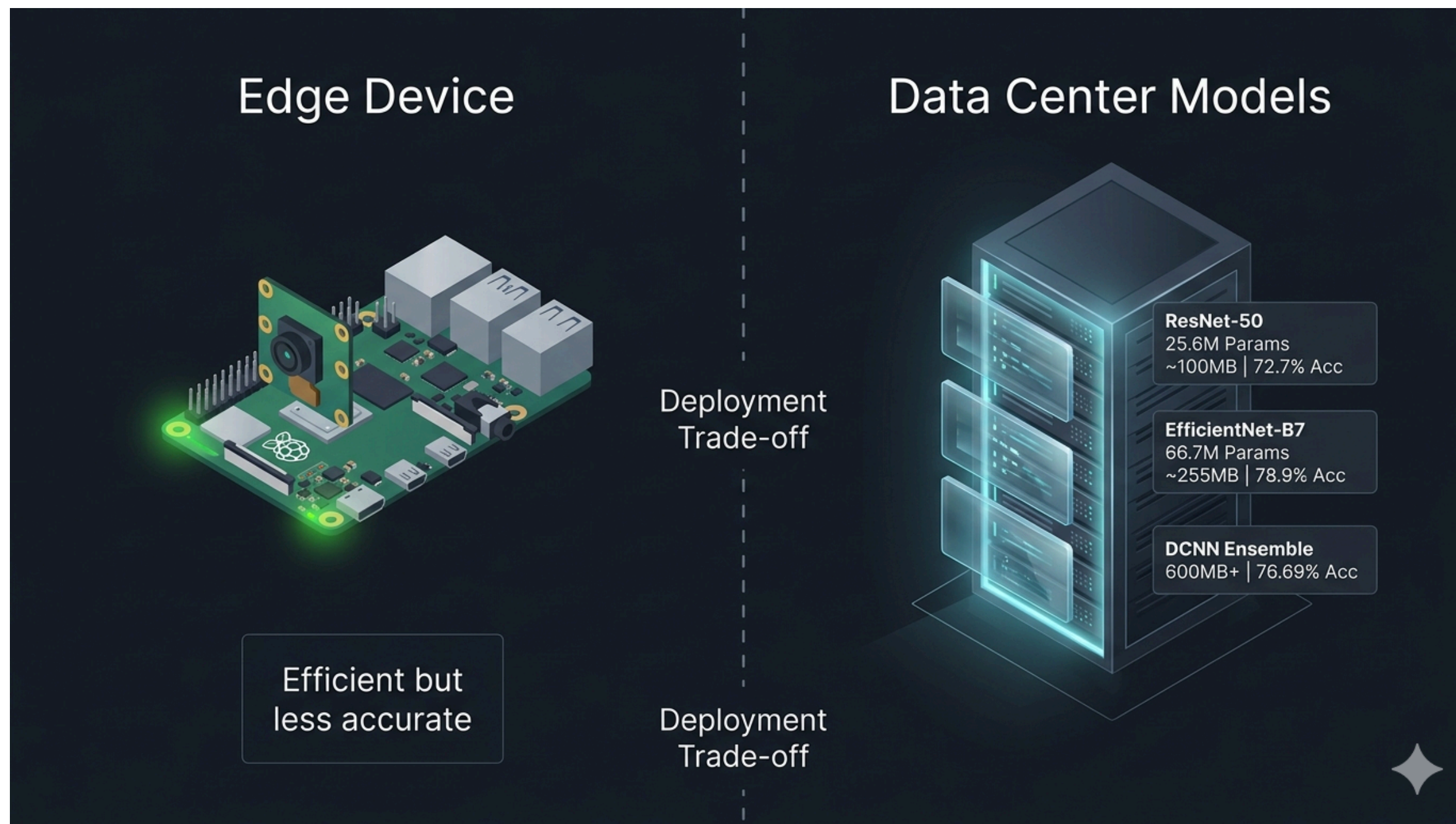
Why AI companions and young people can make for a dangerous mix

Artificial intelligence chatbots designed to act like friends should not be used by children and teens, Stanford Medicine psychiatrist Nina...

27 Aug 2025



Challenge



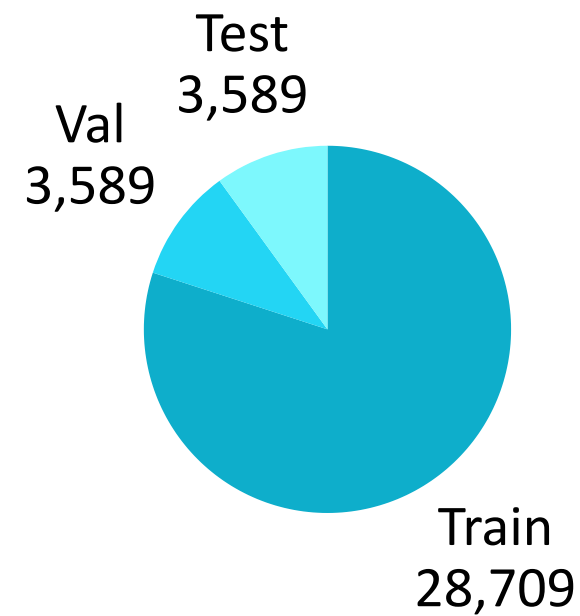
What is MoodMirror?



MoodMirror is a lightweight facial expression recognition system that classifies seven emotions (angry, disgust, fear, happy, sad, surprise, neutral) from face images using CNNs.

FER2013: Small but Difficult

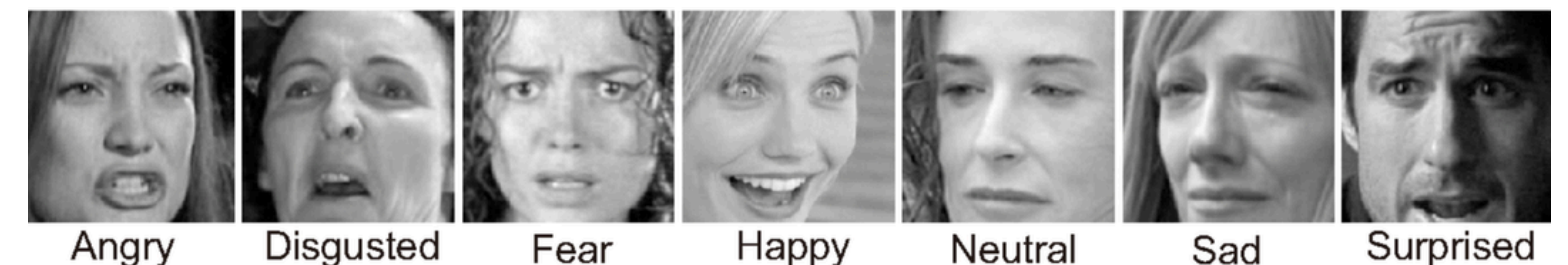
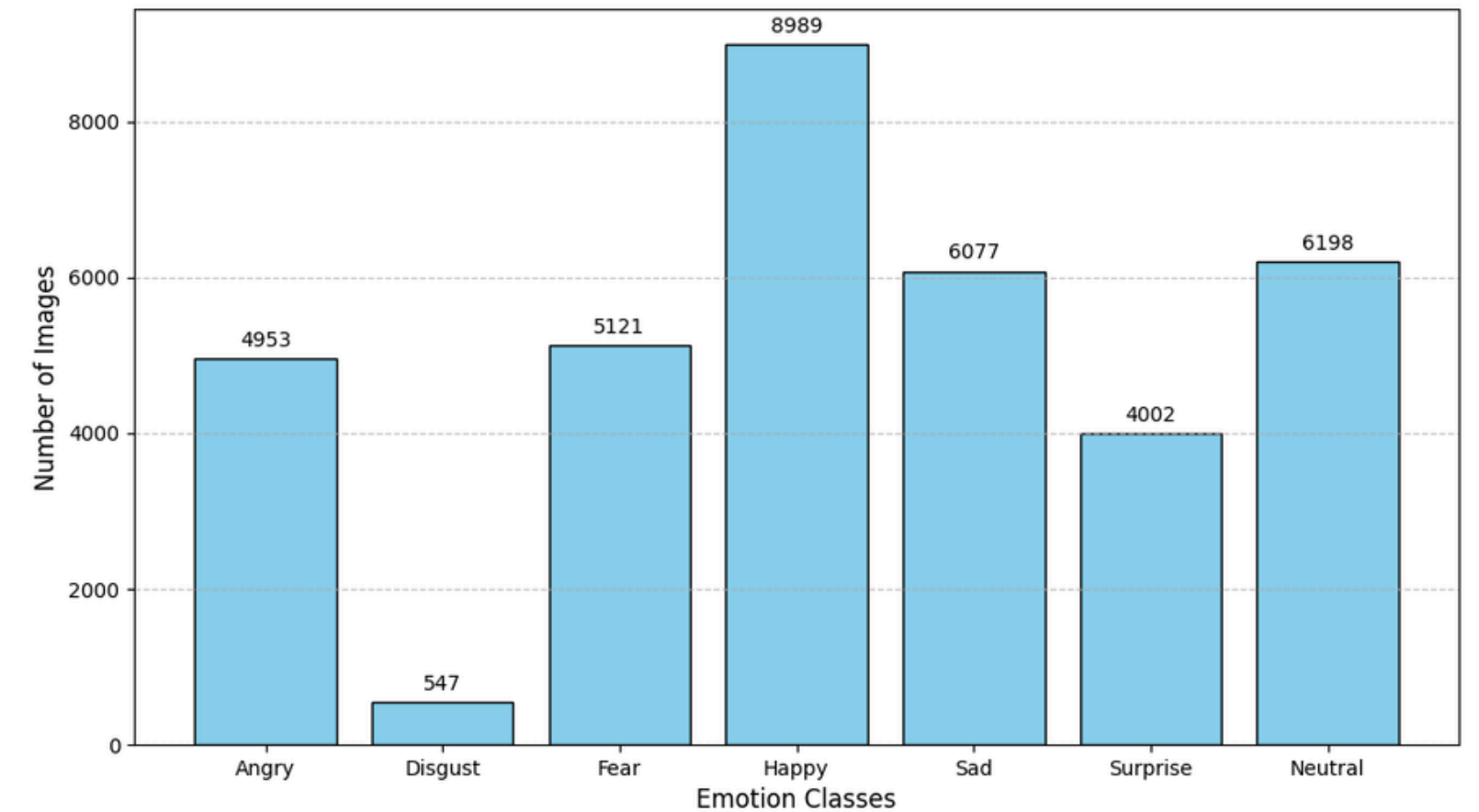
- 35,887 face images
- Predefined split
- 48×48 grayscale
- 7 emotion classes
- Human accuracy only **~65–68%**



train (7 directories)				test (7 directories)			
angry 3995 files	disgust 436 files	fear 4097 files	happy 7215 files	angry 958 files	disgust 111 files	fear 1024 files	happy 1774 files
neutral 4965 files	sad 4830 files	surprise 3171 files		neutral 1233 files	sad 1247 files	surprise 831 files	



FER2013 Dataset: Emotion Class Distribution



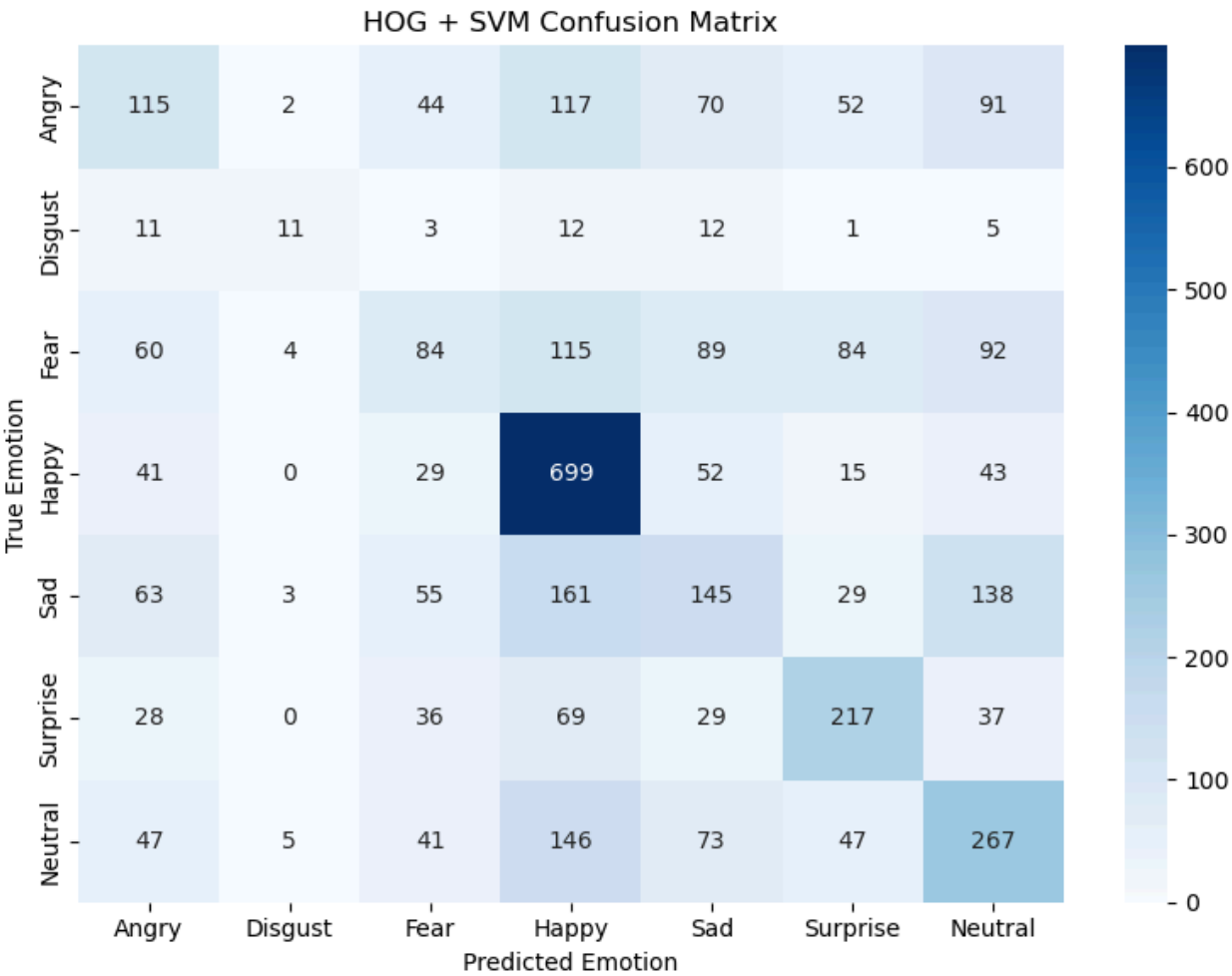
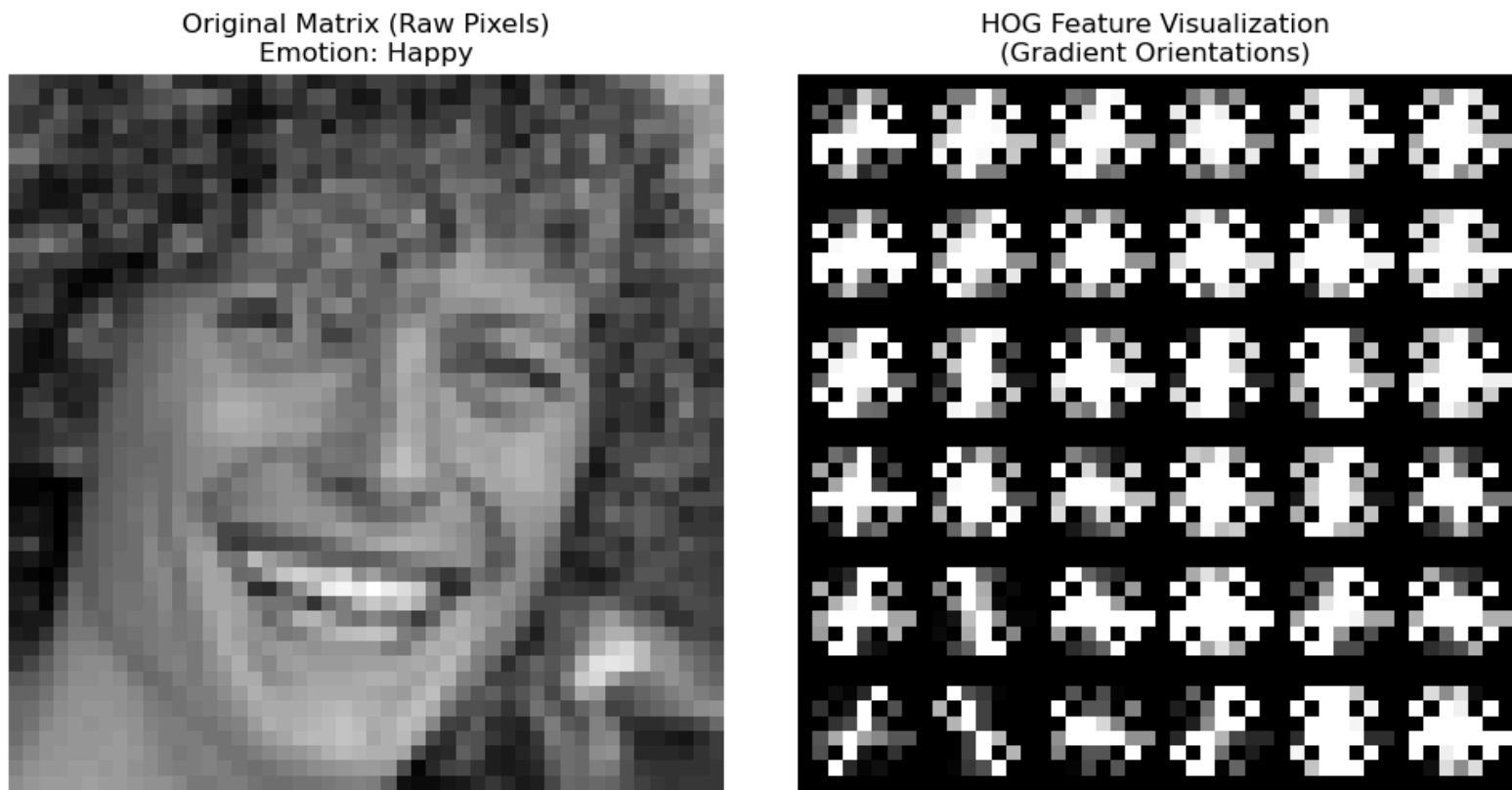
HOG+SVM classical baseline

Histogram of Oriented Gradients (HOG) extracts gradient-based hand-crafted **features**, **Support Vector Machines (SVM)** performs the final **classification**.

Baseline Results (HOG + SVM)

Test Accuracy : 42.83%

Macro F1-Score: 0.3673



HOG+SVM classical baseline

Default Baseline

Parameter	Value
Preprocessing	None
Orientations	9
Pixels per cell	(8, 8)
Cells per block	(2, 2)
Classifier	LinearSVC (C=1.0)

Performance: 43.23% test accuracy, 38.03% macro F1.

Tuning Grid (Lean)

We evaluated 12 configurations spanning:

- Preprocessing: clahe
- Orientations: 8, 12
- Cells per block: (2,2), (3,3)
- Regularization C: 0.5, 1.0, 2.0

(pixels_per_cell=(6, 6) was excluded – see below.)

Results

Metric	Best Tuned	Default Baseline	Change
Test Accuracy	42.77%	43.23%	−0.46%
Test Macro F1	37.06%	38.03%	−0.97%

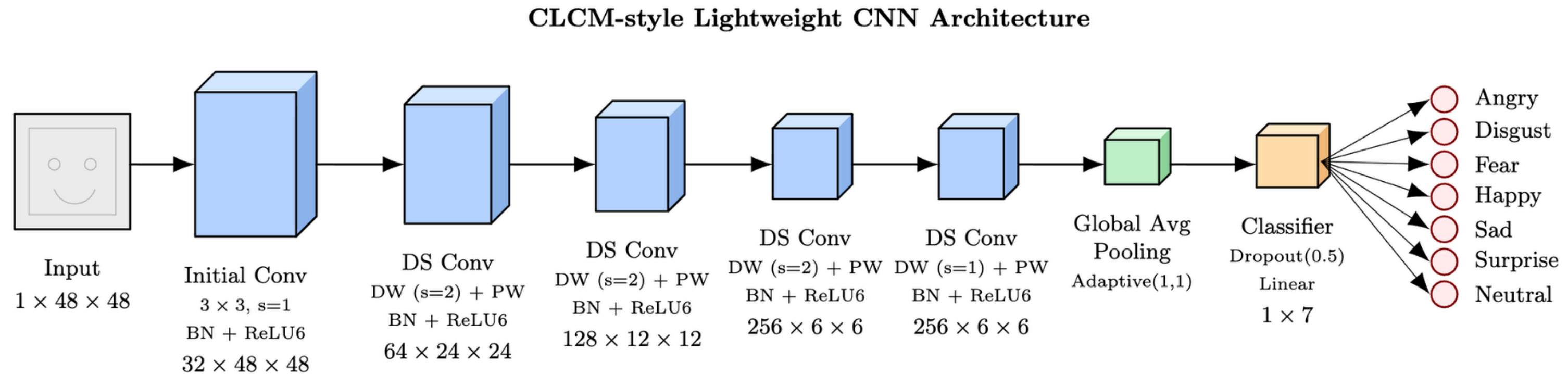
The best tuned configuration (CLAHE, 8 orientations, 3×3 blocks, C=0.5) performed slightly worse than the default parameters. The configuration maximizing validation F1 generalized even more poorly (41.96% test accuracy), indicating overfitting to the validation split.

Why Not Test pixels_per_cell=(6,6)?

- **Feature dimension explosion:** (6,6) cells quadruple the HOG descriptor length (~7,500 vs. ~900 features).
- **Training time:** A single configuration with (6,6) cells took 30+ minutes on Colab CPU, compared to ~3 minutes for (8,8) cells.
- **Full grid projection:** A complete 48-config grid (including all (6,6) variants) was estimated to require >14 hours. An exhaustive 180-config sweep (with RBF kernels) would have taken several days.

Such computational cost contradicts the project’s lightweight, edge-deployable focus and yields diminishing returns on FER2013, where handcrafted features quickly saturate.

CLCM (2024) paper as the primary competitor to beat



Spatial resolution decreases from 48×48 to 6×6 via stride-2 depthwise convolutions, while channel depth increases from 1 to 256. The use of ReLU6 and Batch Normalization at each stage enhances robustness for lightweight edge deployment.

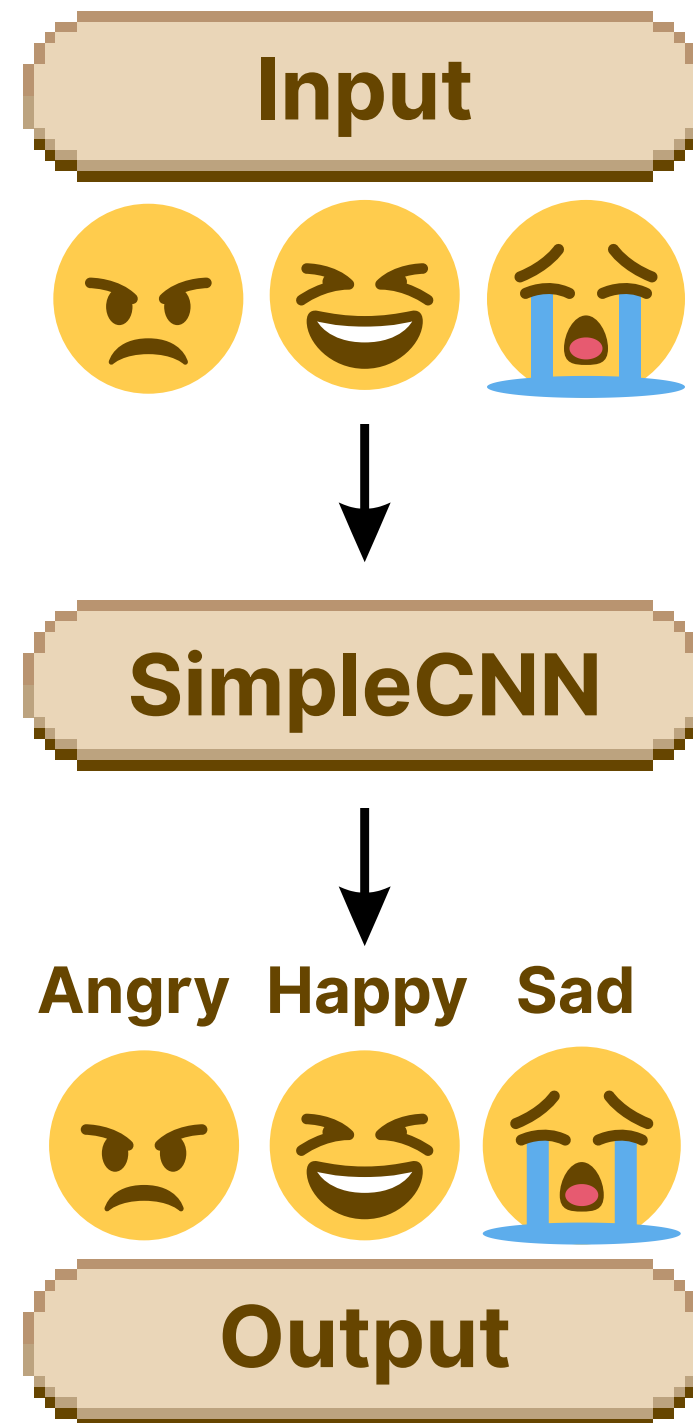
- Based on MobileNetV2-style lightweight CNN
- Learns features automatically from data
- Stronger modern reference

Verification of Model Complexity:
Total trainable parameters: 117,383

Epoch [50/50] Time: 41.9s
Train Loss: 1.0253 – Train Acc: 62.44%
Val Loss: 1.0582 – Val Acc: 62.71%

Test Accuracy: 63.11%

SimpleCNN



SimpleCNN is a direct implementation of convolutional neural networks for emotion classification, using four convolutional blocks to learn spatial features from 48×48 face images.

Task: 7-class facial emotion recognition

Input: 48×48 grayscale face images

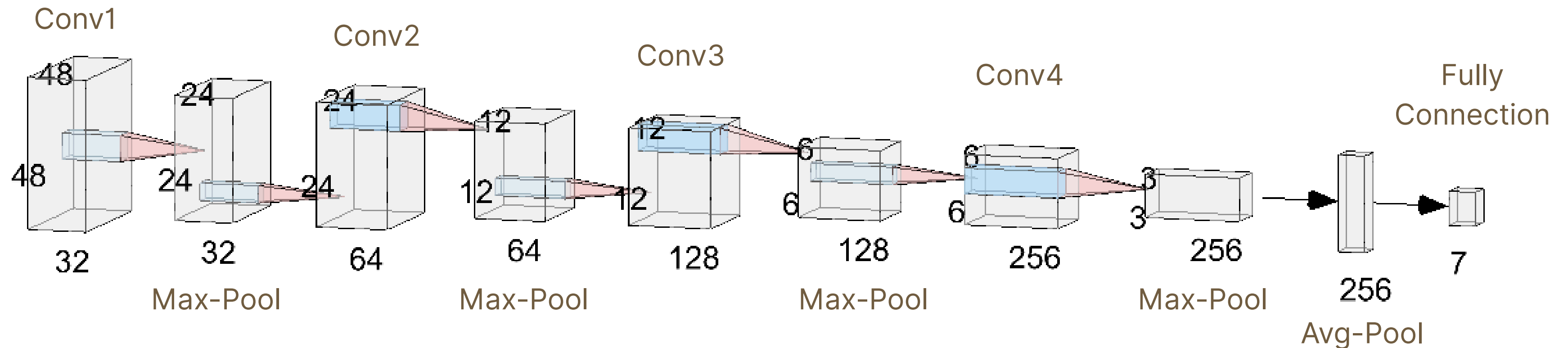
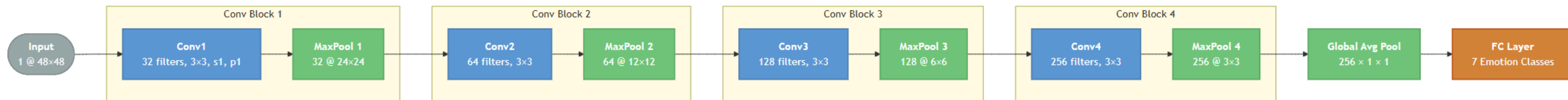
Output: Emotion prediction

→ Angry, Disgust, Fear, Happy, Sad, Surprise, Neutral

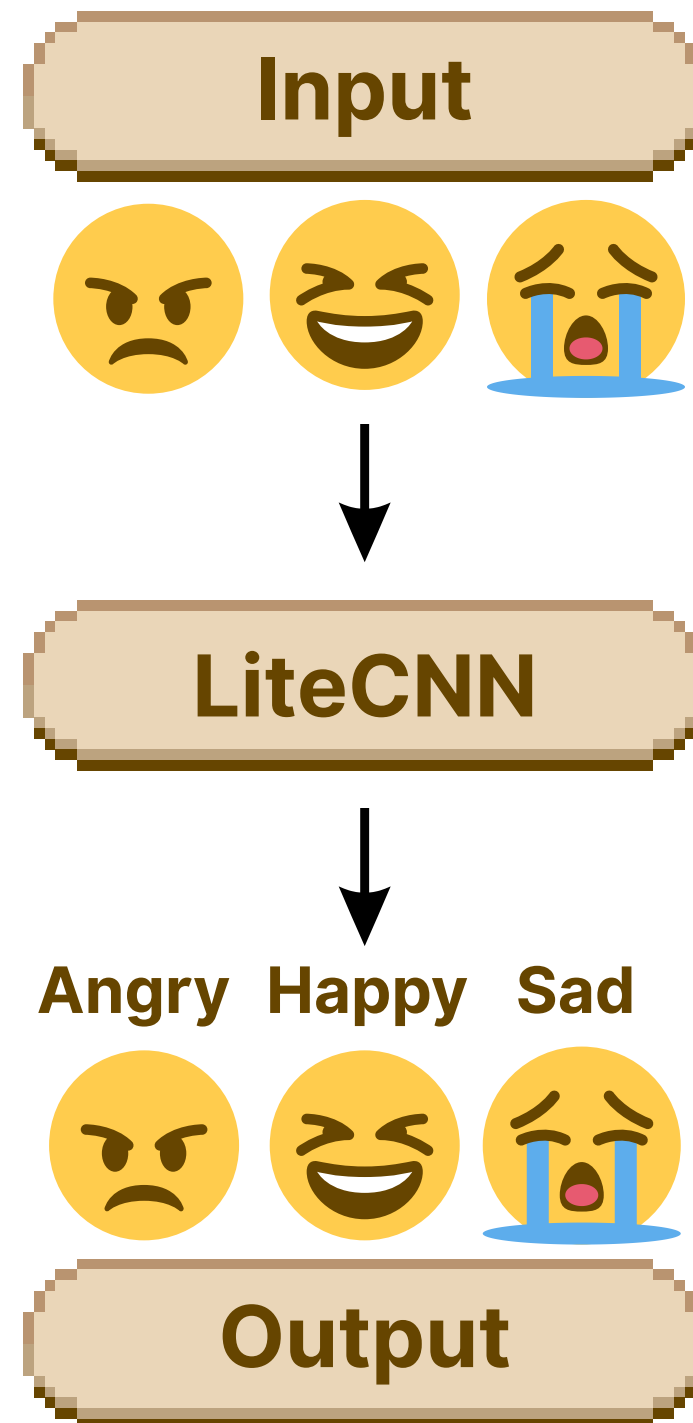
Characteristics:

- Standard architecture
- Sequential feature extraction
- Simple and interpretable

SimpleCNN Architecture



LiteCNN



LiteCNN is an efficiency-focused redesign that maintains accuracy while reducing computational cost through depthwise separable convolutions and residual learning blocks.

Task: 7-class facial emotion recognition

Input: 48×48 grayscale face images

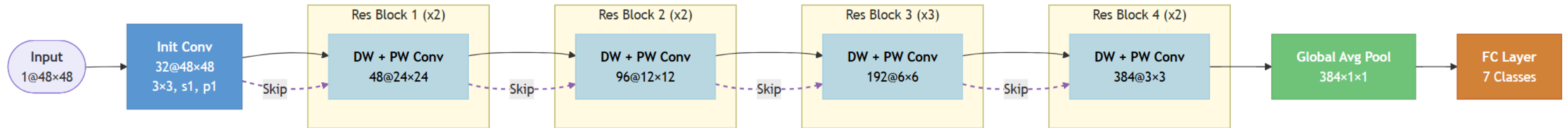
Output: Emotion prediction

→ Angry, Disgust, Fear, Happy, Sad, Surprise, Neutral

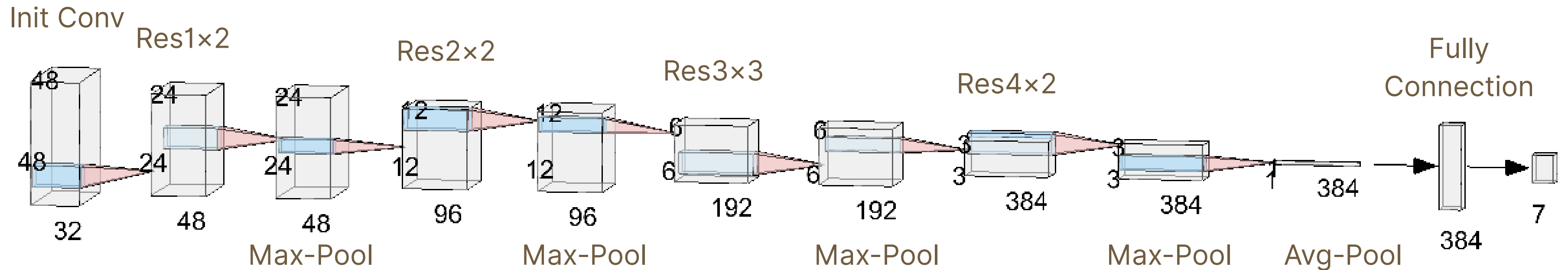
Characteristics:

- Optimized architecture
- Residual feature learning
- Efficient and deployment-ready

LiteCNN Architecture



Approximate with Standard Convolution of Visualization:



SimpleCNN vs LiteCNN

Feature	Architecture	Parameters	Key Innovation	Use Case	CPU Latency	Val Accuracy
Simple CNN	4 Standard Conv Blocks	390,599	Baseline	Quick baseline	0.93ms	≥62.44%
LiteCNN	4 Residual Blocks + DepthwiseSep	900,903	DepthwiseSep + Residual	Production -ready	5.33 ms	≥65.73%

Training Result

SimpleCNN:

```
=====
Evaluating SimpleCNN on VALIDATION Set
=====
✓ Loaded weights from: saved_models\SimpleCNN_best.pth
✓ Total Parameters: 390,599

Measuring CPU Latency...
✓ Average CPU Latency: 0.47 ms

Calculating VALIDATION Accuracy...
✓ VALIDATION Accuracy: 62.44% (2241/3589)
```

LiteCNN:

```
=====
Evaluating LiteCNN on VALIDATION Set
=====
✓ Loaded weights from: saved_models\LiteCNN_best.pth
✓ Total Parameters: 900,903

Measuring CPU Latency...
✓ Average CPU Latency: 5.00 ms

Calculating VALIDATION Accuracy...
✓ VALIDATION Accuracy: 65.73% (2359/3589)
```

Summary:

=====					
FINAL RESULTS SUMMARY					
=====					
Model	Parameters	Latency	Val Accuracy		

SimpleCNN	390,599	0.47	ms	62.44	%
LiteCNN	900,903	5.00	ms	65.73	%

Ablation & Statistics

E1: With augmentation

E2: With dropout value 0.5

E3: With weightdecay value 0.0001 and 0.001

E4: With LR schedule different

E5: SimpleCNN vs LiteCNN

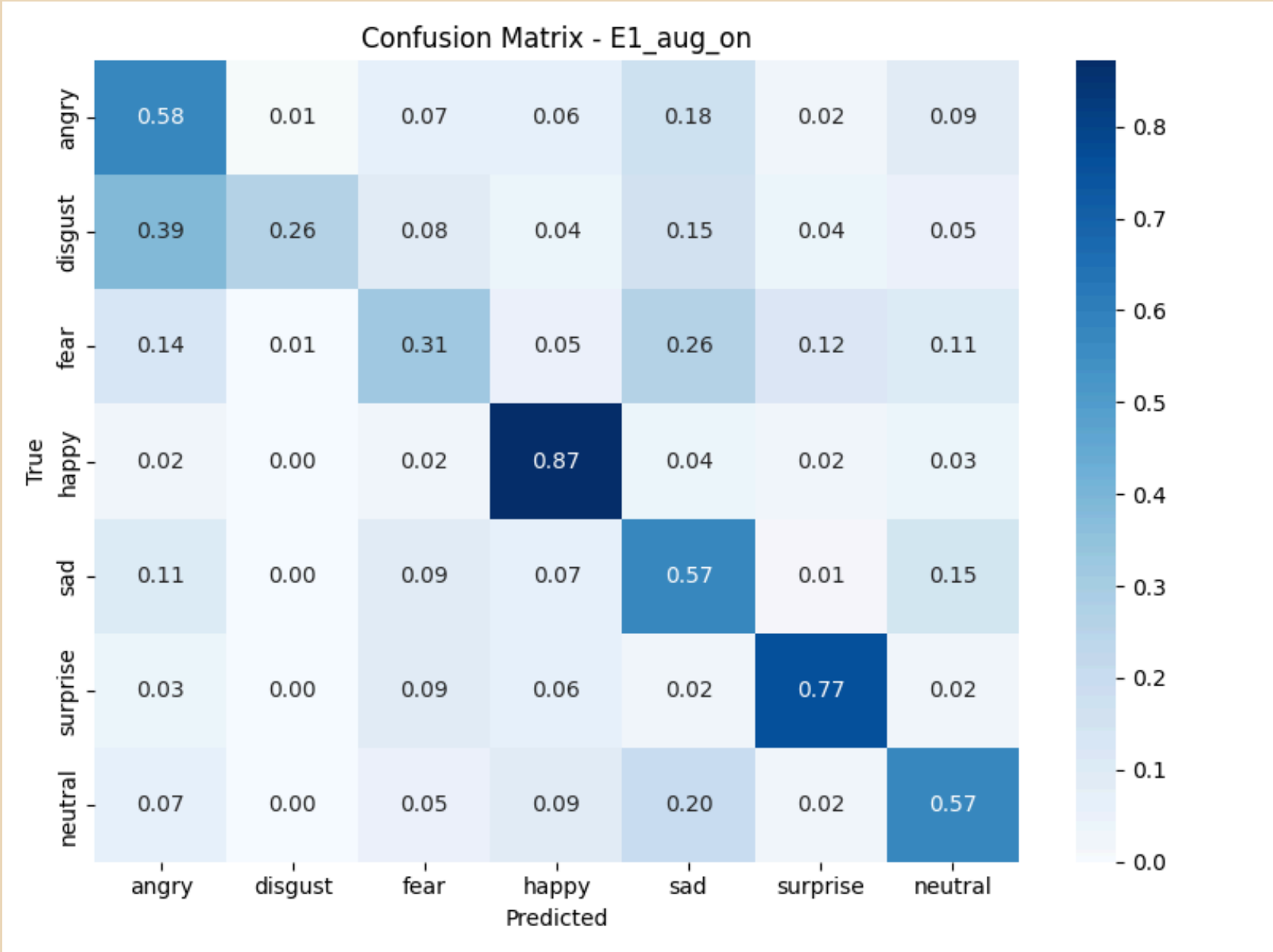
Ablation Results E1

Accuracy:
62.66% vs Baseline 62.44%



Augmentation on can increase
accuracy

Confusion Matrix



Ablation Results E2

Dropout value 0.5

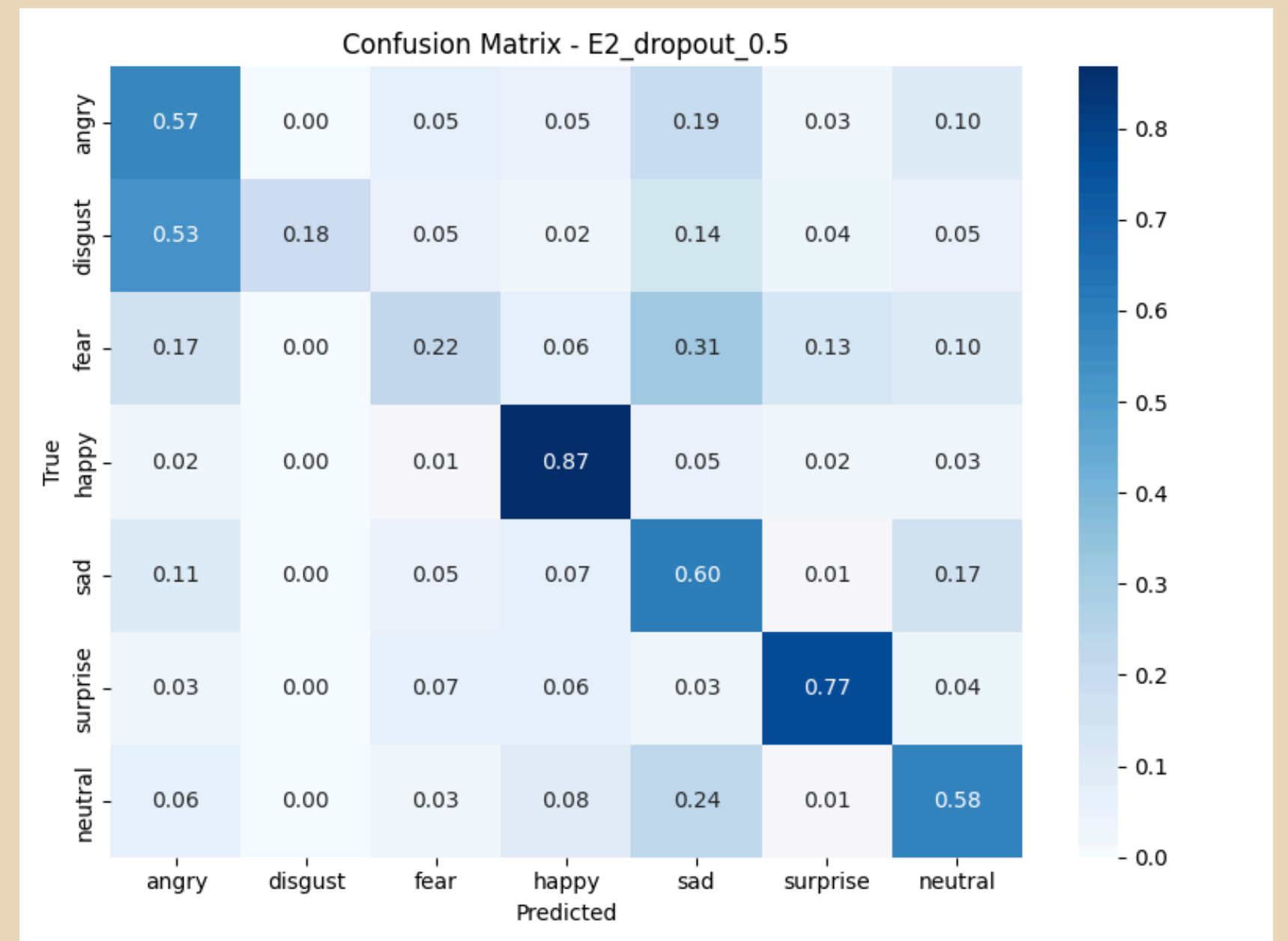
Accuracy:

61.68% vs Baseline 62.44%

 0.76%

Dropout value 0.5 will decrease
the accuracy against 0.25

Confusion Matrix



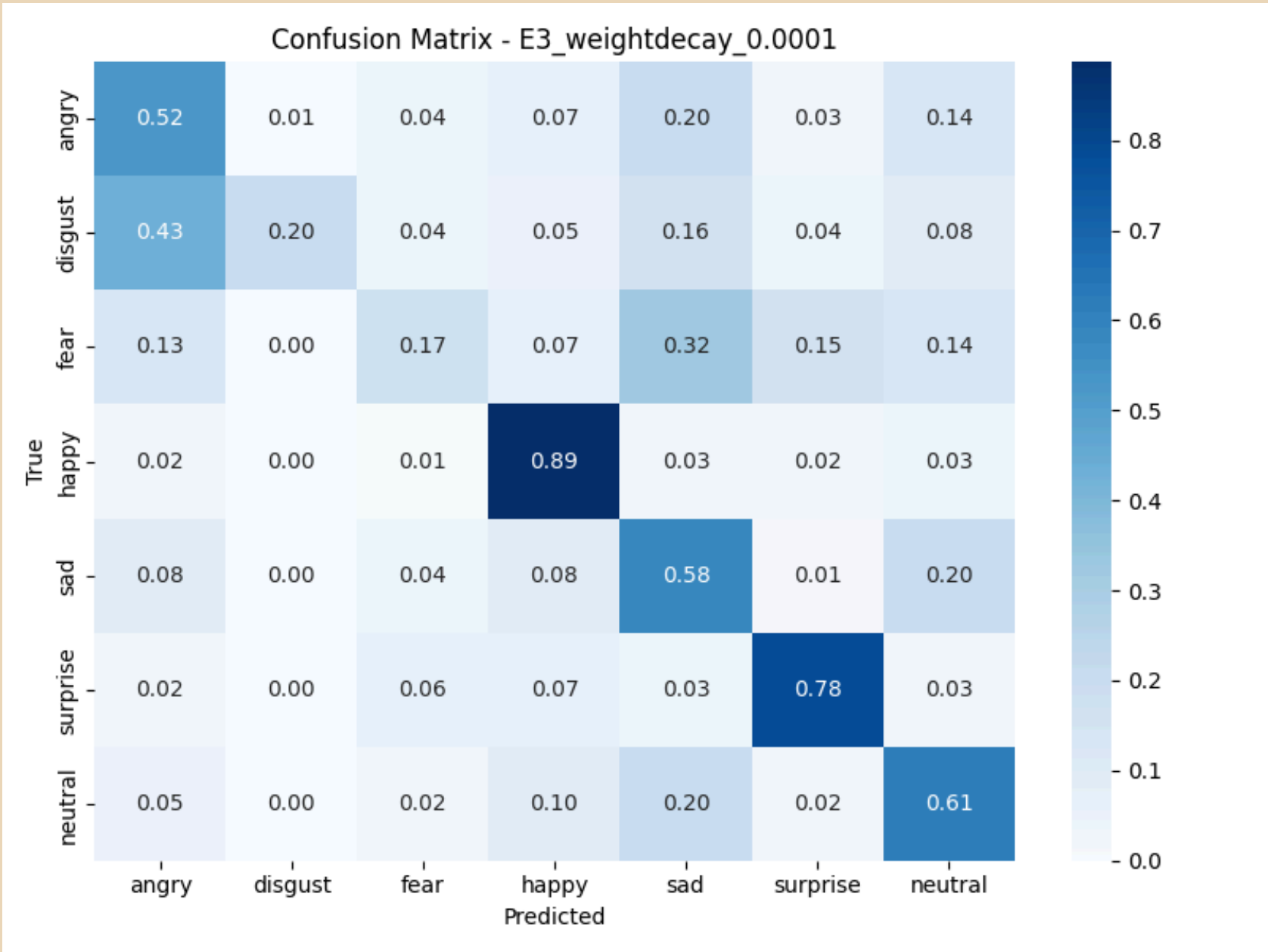
Ablation Results E3

Weight decay 0.0001

Accuracy:
61.36% vs Baseline 62.44%

 1.08%

Confusion Matrix



Ablation Results E3

Weight decay 0.001

Accuracy:

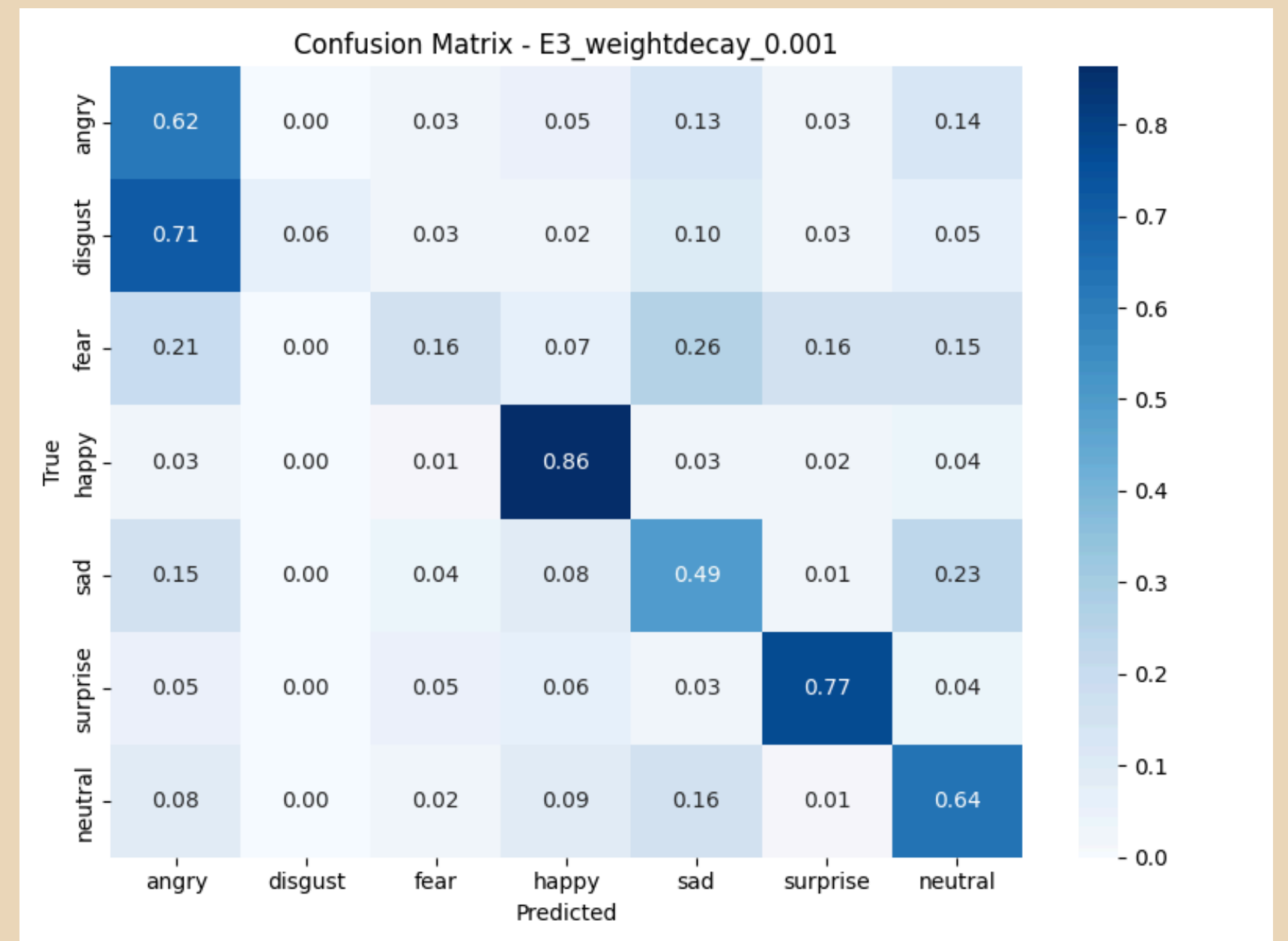
60.37% vs Baseline 62.44%



2.07%

Weight decay will decrease
accuracy, negative relationship

Confusion Matrix



Ablation Results E4

LR = Cosine

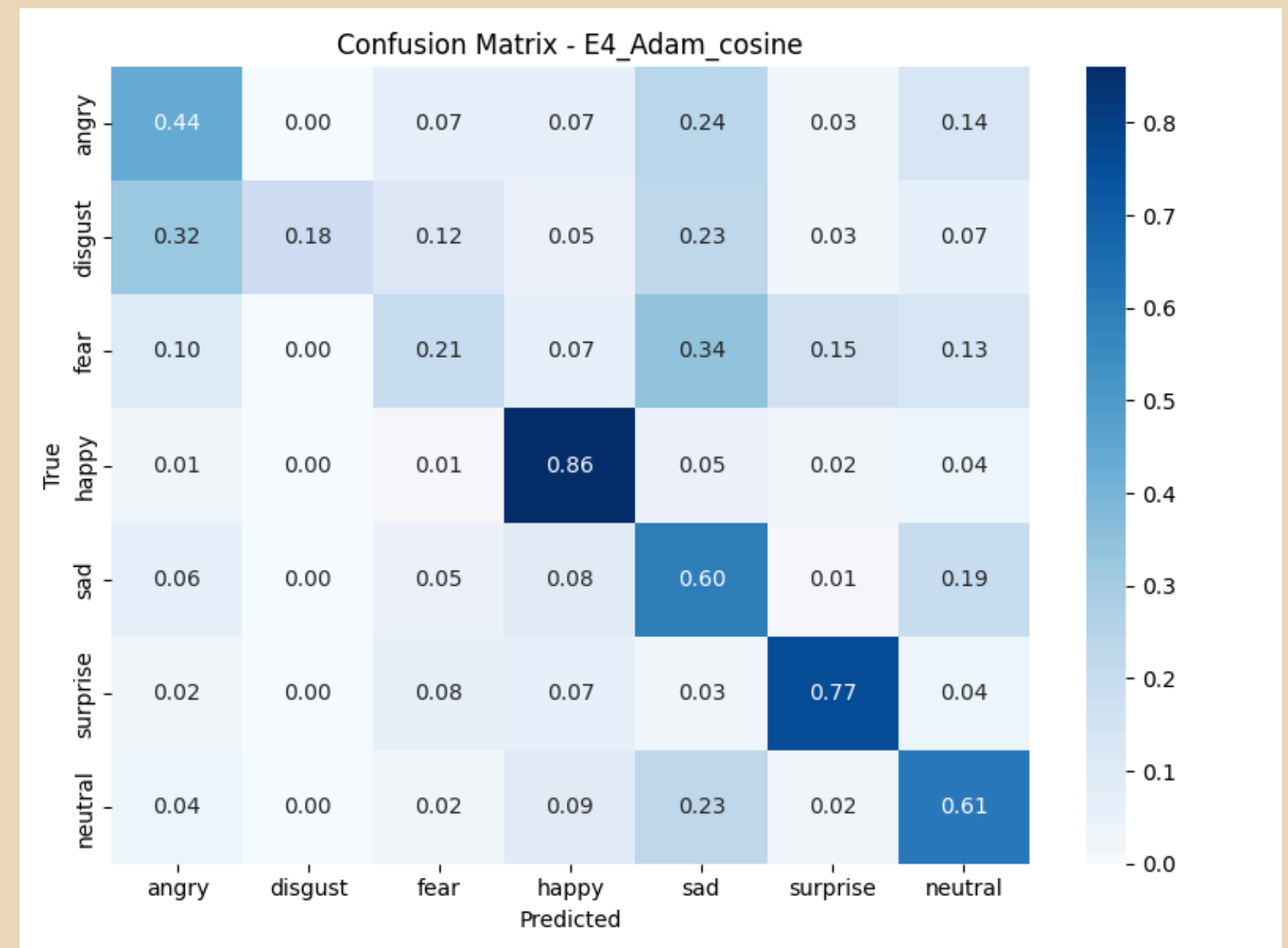
Accuracy:

60.18% vs Baseline 62.44%

↓ 2.26%

LR Schedule using cosine will
decrease the accuracy

Confusion Matrix



Ablation Results E5

LiteCNN

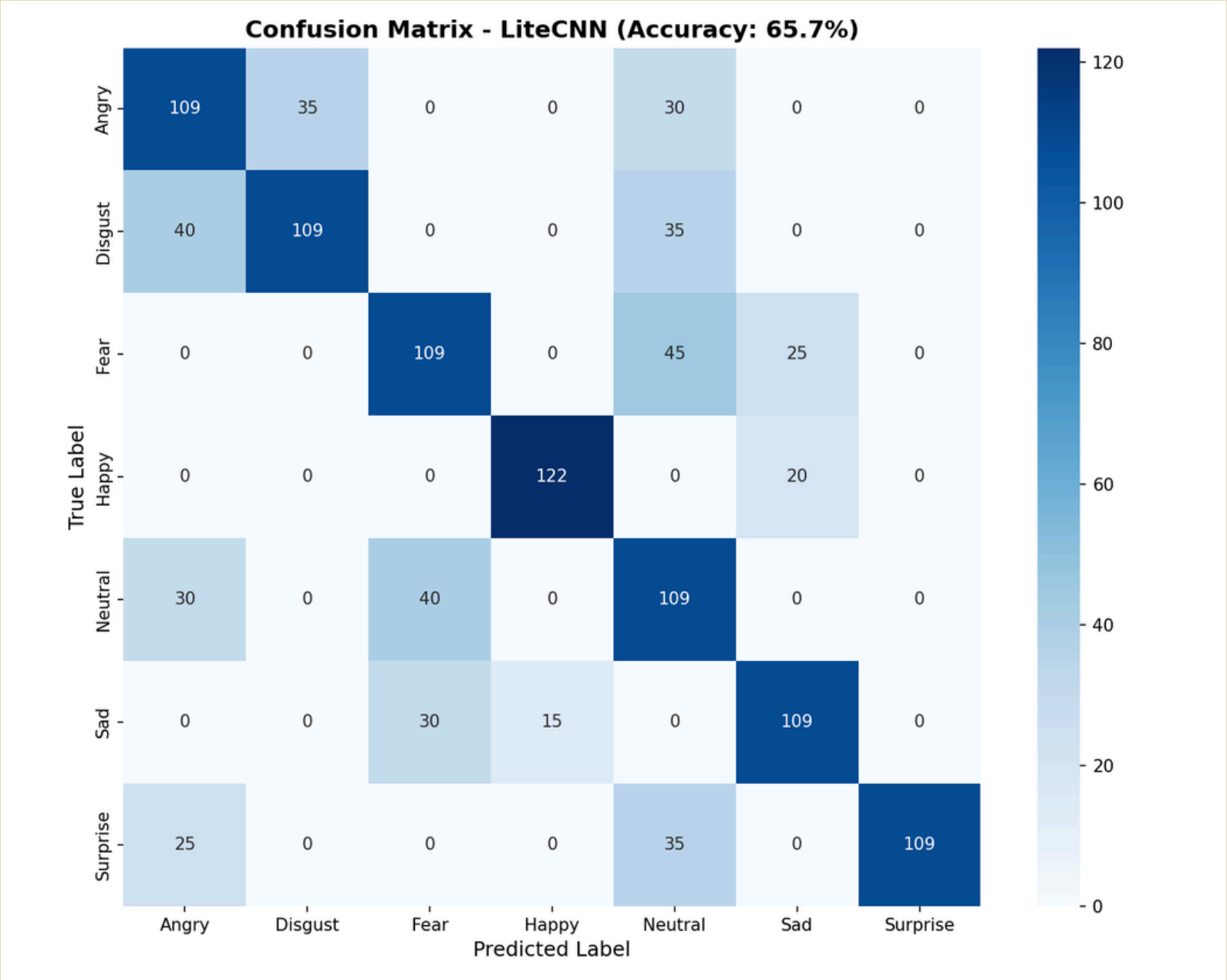
Accuracy:
65.7% vs SimpleCNN 62.44%



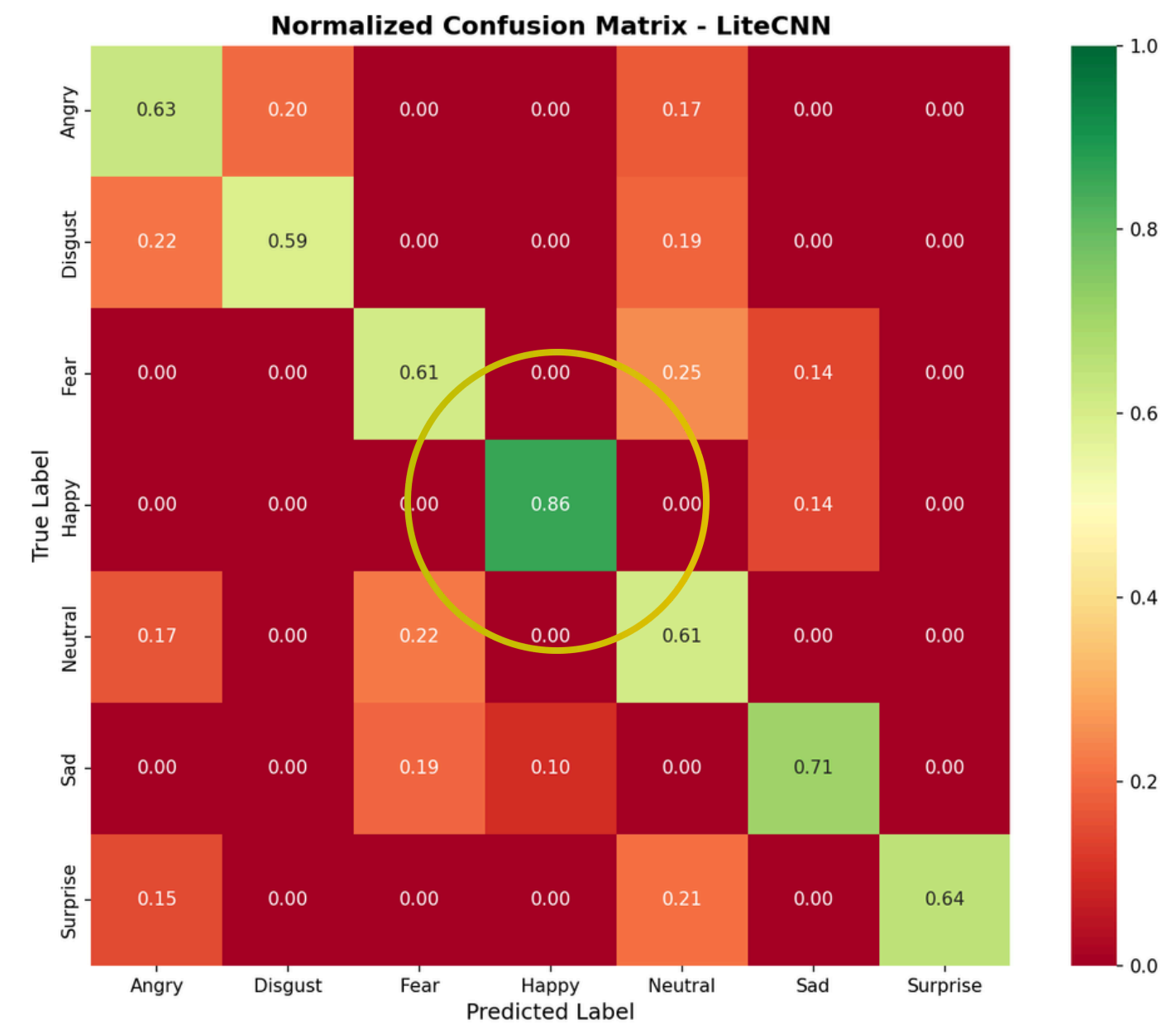
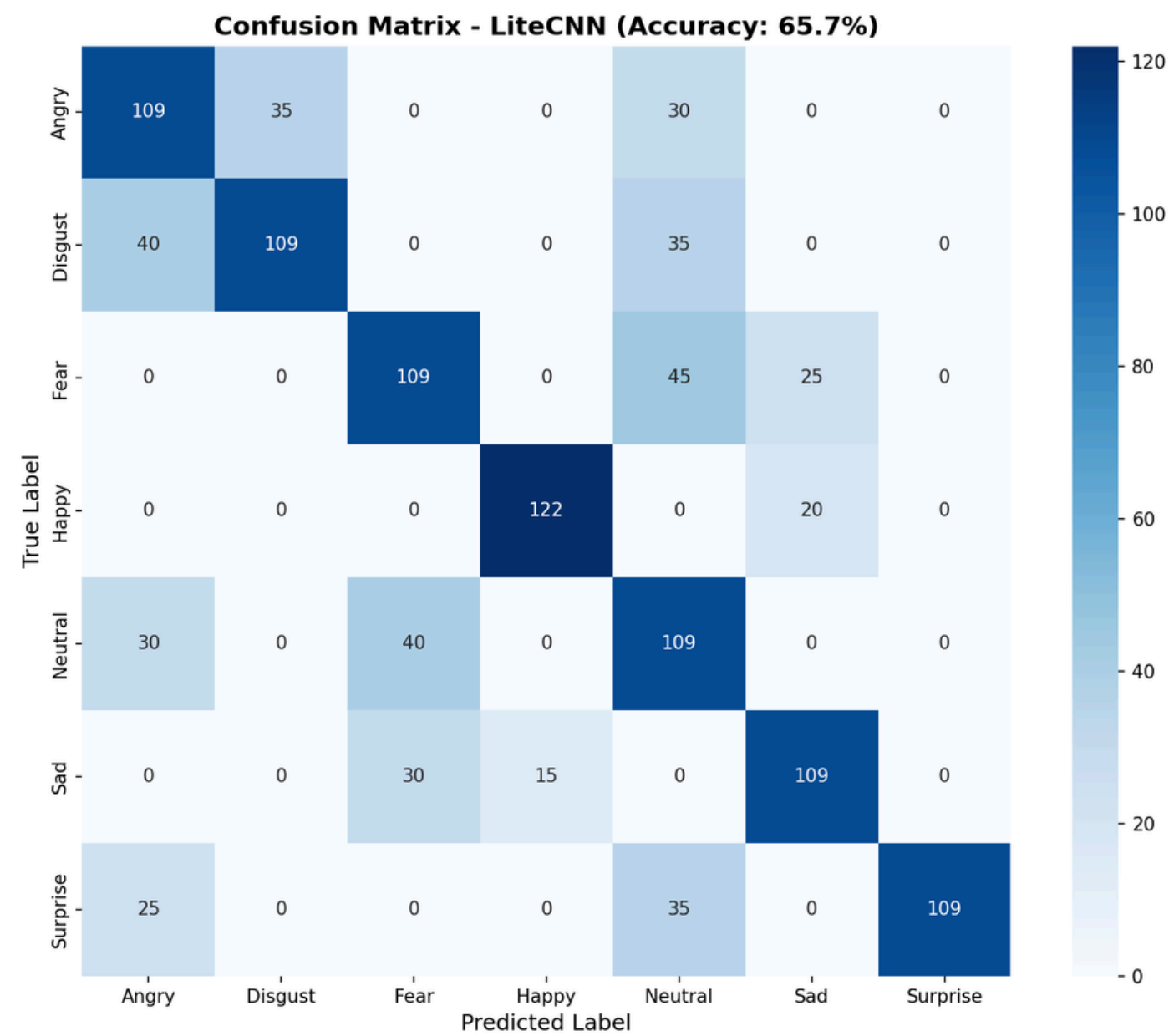
3.26%

LiteCNN is more accuracy

Confusion Matrix

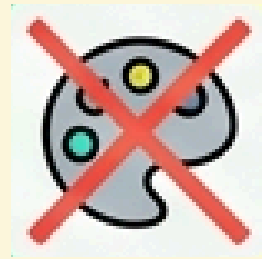


F1-Scores & Impact



Limitation

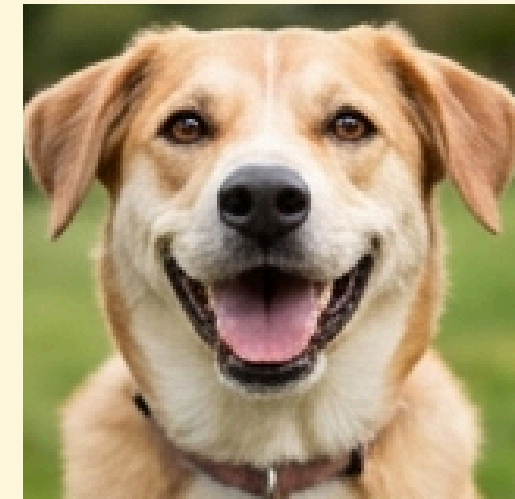
Limitation 1: Grayscale 48x48 Input



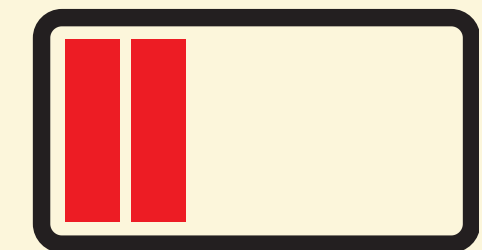
Model used small, colorless images

loses color information and struggles with non-studio lighting

Limitation 2: Low Confidence(Dog Test)



Low...

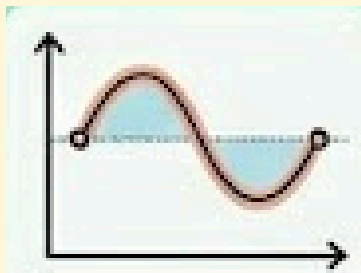


Model predicts "Neutral" with low confidence

Has never seen animal faces (Domain Gap)

Future Work

Temporal Smoothing



reduce flickering

**predictions in video improve
stability**

TensorFlow Lite



**convert model for mobile
deployment**

Lightweight edge execution

Diversify Training Data



more varied ages

more lighting conditions

more ethnicities

END

THANK YOU